

GENOXA

Interactive Bulk RNA-Seq Analysis Platform

Version 0.2

USER MANUAL

A no-code guide for GDC cohort selection, RNA-Seq differential expression, gene ontology over-representation, and machine-learning based gene modeling

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Table of Contents

1. Overview
2. Getting Started: Selecting Your Cohort
3. Setting Analysis Parameters (Left Sidebar)
4. Running the Analysis
5. Results Provided
6. Downloading Plots, Tables, and Data
7. Generating the Analysis Report
8. Troubleshooting & Common Messages
9. About GENOXA

1. Overview

GENOXA is an interactive, no-code web application for analyzing bulk RNA-Seq gene expression data. It combines traditional bioinformatics workflows (limma and edgeR) with machine-learning based gene modeling, so that users without programming experience can go from a case cohort on the NCI Genomic Data Commons (GDC) portal to a finished, shareable analysis report.

GENOXA was selected as a winning tool in the National Cancer Institute (NCI) GDC Analysis Tool Challenge (2025), and integrates directly with TCGA datasets available through the GDC Data Portal.

1.1 Who this manual is for

This manual is written for researchers, students, and analysts who want to run a differential expression and/or machine-learning analysis on RNA-Seq count data sourced from the GDC portal, without needing to write R code. Familiarity with basic RNA-Seq concepts (counts, normalization, fold-change, p-values) is helpful but not required — each parameter is explained in plain language in this manual and inside the app's own "User Manual" tab.

1.2 Key features

- Traditional bioinformatics results for bulk RNA-Seq data using limma and edgeR.
- Interactive, publication-quality visualizations, downloadable as .png or .pdf with customizable size and resolution.
- Multiple color palettes and styling options (titles, axis labels, legends) for every plot.
- Comprehensive Gene Ontology (GO) over-representation analysis, including bar, dot, network, heat, enrichment map, UpSet, and KEGG pathway plots.
- Machine-learning based gene modeling with performance, variable-importance, and ROC plots.
- One-click, parameterized analysis report, downloadable as HTML, PDF, or Word.

1.3 GENOXA workflow at a glance

GENOXA follows a fixed, left-to-right pipeline:

1. Select Data — build a cohort in the GDC Data Portal and import it into GENOXA.
2. Data Preprocessing — filter low-count genes and normalize the count matrix.
3. Differential Expression (DE) Analysis — identify significantly up- and down-regulated genes between two groups.
4. GO Over-Representation Analysis (ORA) — test whether DE genes are enriched for specific biological functions/pathways.
5. Gene Modeling using Machine Learning (ML) — build and compare predictive models using DE genes as features.
6. Results — browse and download every table and plot produced by the steps above.
7. Report — generate a single document summarizing all parameters and results.

Note: All six analysis steps use the parameters set in the left sidebar. Set the parameters for every step before clicking "Start Analysis" at the bottom of the sidebar, because that single button runs the entire pipeline in one pass.

2. Getting Started: Selecting Your Cohort

GENOXA does not accept manual file uploads. Instead, it receives its input cohort directly from the GDC Data Portal's Cohort Builder and "Analysis Center" launch mechanism. This guarantees that every dataset analyzed in GENOXA is compatible with the application's expected format.

2.1 Filters applied automatically

To make sure only compatible files are used, GENOXA automatically restricts every cohort to files matching:

Filter	Value
Sample type	Tumor and Normal
Data category	Transcriptome Profiling
Data type	Gene Expression Quantification
Experimental strategy	RNA-Seq
Workflow type	STAR – Counts
Data format	TSV
Platform	Illumina
Access	Open

Note: You only need to define your cohort (project, demographics, sample type, etc.) in the GDC portal. GENOXA applies all remaining RNA-Seq file filters automatically.

2.2 Step-by-step: building and importing a cohort

1. Open the GDC Data Portal at portal.gdc.cancer.gov.
2. Open the Cohort Builder to define the cohort you want to analyze.
3. Under General, choose a project (for example, TCGA-PRAD) in the Project section.
4. Optionally refine the cohort under Demographic (Race, Vital Status, Gender, Age, or other clinical variables).
5. Once satisfied with the cohort, go to Analysis Center and select the GENOXA application.
6. Click Play to launch GENOXA with your cohort already attached.
7. Inside GENOXA, click Select Data in the left sidebar.
8. GENOXA automatically imports the cases you selected in the Cohort Builder and shows a live Cohort Summary (cases / files available, and how many will be downloaded).
9. Set Max Samples (files) to Download and, optionally, Max Genes to Keep, then click Download Data.

Note: Files are retrieved from the GDC API in automatic batches of 140. Large cohorts may take several minutes and will download in multiple batches; progress is reported live in the status box below the Download Data button.

2.3 What happens during download

When you click Download Data, GENOXA performs the following steps automatically, logging progress to the on-screen status box:

- Step 1 — Queries GDC for the file manifest matching your cohort and filters.

- Step 2 — Fetches the clinical/demographic fields you selected in the Cohort Builder for every case.
- Step 3 — Downloads gene count files (STAR – Counts TSVs) in batches and parses each one.
- Step 4 — Assembles a single raw gene-count matrix (genes as rows, samples as columns), together with a sample/group metadata table and a clinical/annotation table.

If a cohort is heavily imbalanced (fewer than 2 samples in the Tumor or Normal group) or the sample type was not specified in the cohort builder, GENOXA stops and reports a clear error so you can rebuild the cohort with adequate group sizes before proceeding.

3. Setting Analysis Parameters (Left Sidebar)

Once your cohort has downloaded, use the remaining sidebar sections to configure each analysis step. Every parameter has an equivalent, more detailed explanation inside the app itself, under the User Manual tab. Parameters are grouped by analysis stage:

3.1 Data Preprocessing

Parameter	What it does
Filter raw reads by rowSum	Removes genes whose total read count across all samples falls below this threshold, before differential expression testing (default 5).
Filter raw reads by FilterByExpr	Uses edgeR's filterByExpr() to remove genes that are not consistently expressed across samples, improving the robustness of downstream DE testing (default 5).
Normalization method (calcNormFactors)	Chooses how library-size and composition differences between samples are corrected: TMM, RLE, upperquartile, or none.

Normalization method choices

- **TMM (Trimmed Mean of M-values):** corrects for library-size and compositional differences; best default choice for most RNA-Seq datasets.
- **RLE (Relative Log Expression):** scales samples to a common median relative log expression; assumes most genes are not differentially expressed.
- **Upper Quartile:** scales by the upper quartile of counts; less sensitive to a few highly-expressed genes.
- **None:** no normalization; use only if the data was already normalized elsewhere.

3.2 Differential Expression (DE) Analysis

Parameter	What it does
Reference group for DE	The baseline/control condition (e.g., Normal) that all other conditions are compared against, set via edgeR's releve1().
P-value / adjusted p-value cutoff	Maximum p-value (raw) or adjusted p-value considered statistically significant (default 0.05).
Minimum absolute log2FC	Smallest fold-change magnitude required for a gene to be called differentially expressed (default 1, i.e. $\log_2FC \geq 1$ or ≤ -1).
Adjustment method	Multiple-testing correction applied to p-values: none, bonferroni, holm, BH, fdr, or BY.

3.3 GO Over-Representation (ORA)

Parameter	What it does
Organism database	Human (org.Hs.eg.db) or Mouse (org.Mm.eg.db) gene annotation source.
Key type	Identifier format of your genes (ENSEMBL, SYMBOL, ENTREZID, ALIAS, REFSEQ, UNIPROT, etc.).
Subontology	BP (Biological Process), MF (Molecular Function), CC (Cellular Component), or ALL.
Minimum / Maximum Gene Set Size	Filters out GO terms whose gene sets are too small or too large to be informative (defaults 10 / 500).
P-value / q-value cutoff	Significance thresholds for enriched GO terms (defaults 0.05 / 0.02).

Parameter	What it does
Adjust method	Multiple-testing correction for GO term p-values: none, holm, hochberg, hommel, bonferroni, BH, BY, or fdr.

3.4 Gene Modeling using Machine Learning

Parameter	What it does
Training data set size	Proportion of samples used for training vs. testing (default 0.80, i.e. 80% train / 20% test).
Seed	Random seed for the train/test split, ensuring reproducible results.
Resampling method	How the training data is validated during model fitting: none, boot, boot632, optimism_boot, boot_all, cv, repeatedcv, LOOCV, LGOCV, oob, timeslice, adaptive_cv, adaptive_boot, or adaptive_LGOCV (default repeatedcv).
Number of folds (repeatedcv only)	How many complete sets of cross-validation folds to compute (default 3).
Number of resampling iterations	How many times the model is trained/validated across data subsets (default 10).
Random number seed	Seed used for model training/validation, for reproducibility (default 351).
ML algorithms	The specific algorithms to train and compare (see 3.5). Select a minimum of two and a maximum of four.

Note: GENOXA enforces the two-to-four algorithm rule automatically: selecting fewer than two or more than four models triggers an on-screen warning, and the analysis will not start until the selection is corrected.

3.5 Available ML algorithms

Algorithm	Description
Naive Bayes (naive_bayes)	Probabilistic classifier assuming conditional independence between features; fast and effective for high-dimensional data.
Flexible Discriminant Analysis (fda)	Extension of Linear Discriminant Analysis using non-linear transformations for complex class boundaries.
Support Vector Machine (svmRadial)	Finds an optimal separating boundary using a radial basis function kernel; models complex non-linear relationships.
Boosted GLM (glmboost)	Boosting-based Generalized Linear Model that combines weak learners to reduce overfitting.
Random Forest (rf)	Ensemble of decision trees; robust to overfitting and provides variable-importance estimates.
C5.0	Advanced, interpretable decision-tree algorithm supporting boosting and rule-based models.
glmnet	Regularized GLM using Lasso (L1), Ridge (L2), or Elastic Net penalties for feature selection on high-dimensional data.
Partial Least Squares (kernelpls)	Dimensionality-reduction regression technique suited to highly correlated predictors.

Important Disclaimer: machine-learning results are generated by automated statistical models and should be validated using additional independent methods before any conclusions are drawn. The developers do not take responsibility for claims or decisions made solely on the basis of ML results produced by this application.

4. Running the Analysis

After the cohort has been downloaded and every parameter in the sidebar has been reviewed, click Start Analysis at the bottom of the sidebar. GENOXA will:

- Automatically switch the view to the Results tab.
- Run preprocessing, normalization, DE testing, GO/KEGG over-representation, and ML gene modeling in sequence, using parallel processing where possible.
- Display a progress alert while the analysis runs; large cohorts and multiple ML algorithms will take longer.

Note: Every sample group must contain at least 2 samples. If a group is too small, GENOXA stops and displays an explanatory error instead of producing misleading DE, ROC, or ML results — rebuild the cohort with more samples of the affected tissue type and try again.

5. Results Provided

The Results tab is organized into six sub-tabs, matching the GENOXA workflow. Each plot includes on-screen customization controls (titles, axis labels, legend, color palette) and a separate set of options — size, resolution (dpi), and file type (PNG or PDF) — that apply only to the downloaded copy of the plot.

5.1 User Data Information

Output	Description
Raw Read Counts Data	Interactive table of the raw gene-count matrix, with row/column counts displayed.
Sample and Group Information	Table of sample IDs and their assigned group/tissue type.
Raw Read Counts Plot	Boxplot, column, or violin plot of counts (raw or log2) per sample, with a selectable sample range.
Clinical/Annotation Data	Table of the clinical fields imported from the GDC cohort filter (e.g., demographic variables).

5.2 QC and Data Normalization

Output	Description
Filtered Read Counts Plot	Distribution of counts after the rowSum filter is applied.
FilterByExpr Plot	Distribution of counts after edgeR's filterByExpr() filter is applied.
Before and After Filtering Plot	Side-by-side density plots (log-CPM) comparing raw vs. filtered data.
Data Normalization Plot	Distribution of counts after calcNormFactors() normalization.

5.3 Exploratory Data Analysis

Output	Description
PCA Plot	Principal Component Analysis scatter plot (PC1 vs. PC2), colored by group, to reveal overall sample structure.
MDS Plot	Multi-Dimensional Scaling plot (leading log-fold-change dimensions), colored by group.
Voom Plot	limma-voom mean-variance trend plot, assessing whether the data is ready for linear modeling.
Mean-Variance Trend Plot	Final-model mean-variance relationship after model fitting (plotSA).

5.4 DE Results

Output	Description
DE Table	Full limma/edgeR differential-expression results (logFC, AveExpr, t, P.Value, adj.P.Val, B), with an info box summarizing total, up-, and down-regulated gene counts. Downloadable as: all results, top N hits, up+down regulated genes, up-regulated only, or down-regulated only.
Volcano Plot	log2 fold-change vs. significance ($-\log_{10} p$ or adjusted p), with optional threshold lines and top-gene labels.
DE Heat Map Plot	Clustered heat map of selected DE genes across selected samples, with optional clinical/annotation tracks.

5.5 Gene Modeling (ML) Results

Output	Description
Performance Plot	Comparative performance metrics (e.g., accuracy/kappa) across all trained models and resampling iterations, plus a downloadable performance matrix (CSV).

Output	Description
Variable Importance Plot	Bar or lollipop plot ranking the top predictive genes for each trained model.
ROC Plot	Receiver Operating Characteristic curves for each model, with a downloadable confusion matrix (CSV).

5.6 ORA (Gene Ontology / KEGG) Results

Output	Description
GO over-representation table	enrichGO() results table for the top N DE genes (N is user-configurable), downloadable as CSV.
Bar Plot / Dot Plot	Ranked GO term enrichment visualizations; the dot plot can be split by ontology (BP/MF/CC) when Subontology = ALL.
Gene-Concept Network Plot	Network linking enriched GO terms to their associated genes (cnetplot).
Heat Plot	Gene-by-GO-term heat map of enrichment results.
Enrichment Map Plot	Network of enriched GO terms connected by shared-gene similarity (emapplot).
UpSet Plot	Set-intersection plot showing gene overlap across enriched GO terms.
KEGG Results & Plot	enrichKEGG() results table plus an interactive Pathview diagram for a selected KEGG pathway ID, downloadable as PNG or PDF.

6. Downloading Plots, Tables, and Data

Nearly every plot in GENOXA has its own Download Plot button, alongside independent controls for:

- Height and width of the exported image (in inches).
- Resolution (dpi).
- File type — PNG or PDF.

These download settings apply only to the exported file; they do not change how the plot appears on screen. Data tables (DE results, GO/KEGG results, performance matrices, confusion matrices) are downloadable as CSV files from buttons located directly beneath each table.

7. Generating the Analysis Report

The Report tab produces a single, self-contained document summarizing the methods, parameters, and key results of your analysis (raw and filtered data dimensions, normalization method, DE thresholds, GO/KEGG settings, and ML configuration).

7.1 How to generate a report

10. Go to the Report tab (final item under Results).
11. Choose a Report Format: PDF, Word, or HTML.
12. Click Download Data analysis Report.

Note: The report captures the exact parameter values used for the run that is currently on screen. If you change a parameter and re-run the analysis, download a new report to keep the two in sync.

8. Troubleshooting & Common Messages

Message / Symptom	Likely cause and fix
"No cohort received from the portal."	GENOXA was opened directly rather than launched from the GDC Cohort Builder's Analysis Center. Relaunch GENOXA from the GDC portal after building a cohort.
"Please select the sample type from the cohort builder."	The GDC cohort filter did not include a tissue_type / sample_type facet. Add a Tumor/Normal sample-type filter in the Cohort Builder and relaunch.
"Cohort is too imbalanced" / group counts error	One group (e.g., Normal) has fewer than 2 samples. Rebuild the cohort to include more samples of that tissue type.
"0 significant DE genes" warning	No genes passed the current p-value/adjusted p-value or log2FC thresholds. Loosen the DE cutoff or log2FC threshold in the sidebar and re-run.
No heat map / performance / ROC plot shown	Too few significant DE genes, or fewer than 2 ML models trained successfully; usually resolved the same way as the item above.
ML minimum/maximum model warning	Fewer than 2 or more than 4 ML algorithms were selected. Adjust the selection in the Gene Modeling section of the sidebar.
Sample/label mismatch error	An internal ordering mismatch between the count matrix and sample metadata was detected before analysis; rebuild the cohort and re-run rather than trusting DE/ML results from that run.

9. About GENOXA

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9.1 News and updates

- February 26, 2026 — GENOXA Ver 0.2 selected as a winning tool in the National Cancer Institute (NCI) Genomic Data Commons (GDC) Analysis Tool Challenge (2025), recognizing its innovation in integrating advanced analytical tools with GDC infrastructure.
- May 8, 2025 — GENOXA Ver 0.2 released, featuring integration with TCGA datasets from the GDC portal as part of the GDC Tool Challenge submission.
- November 25, 2024 — GENOXA Ver 0.1 released.